

24-Bit Analog-to-Digital Converter (ADC) with Built-in Temperature Sensor

FEATURES

- On-chip temperature measurement
- On-chip active low noise amplifier with a gain of 128
- On-chip oscillator requiring no external component
- On-chip power-on-reset
- Simple digital control and serial interface: pin-driven controls, no programming needed
- Selectable 10SPS or 40SPS output data rate
- Simultaneous 50 and 60Hz supply rejection
- Current consumption:
 - normal operation $< 1.2\text{mA}$, power down $< 1\mu\text{A}$
- Operation supply voltage range: $2.6 \sim 5.5\text{V}$
- Operation temperature range: $-40 \sim +85^{\circ}\text{C}$
- 8 pin SOP-8 or DIP-8 package

normal operation < 1.2mA, power down < 1uA

On-chip power-on-reset circuitry simplifies digital interface initialization. There is no programming needed for the internal registers. All controls to the HX710 are through the pins.

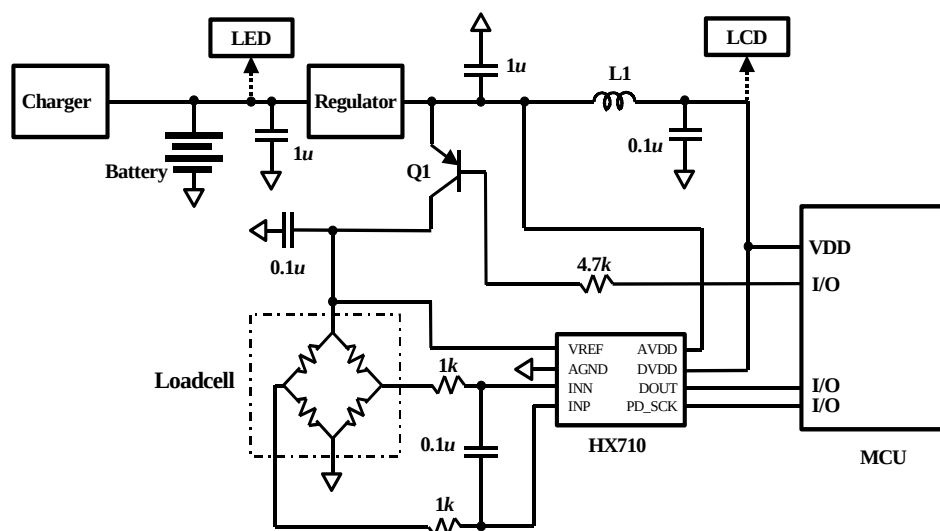


Fig. 1 Typical weigh scale application block diagram

Pin Description

VREF
Ground Input AGND

SOP-8 Package

KEY ELECTRICAL CHARACTERISTICS

Parameter	Notes	MIN	TYP	MAX	UNIT
Full scale differential input voltage range	V(inp)-V(inn)	$\pm 3.9 \cdot V_{REF}$			mV
Common mode input range		AGND+0.9		AVDD-1.3	V
VREF input voltage range		1.8		AVDD	
Output data rate			10/80		Hz
Output data coding	2's complement	800000		7FFFFFFF	HEX
Output settling time ⁽¹⁾			400/50		ms
Input offset drift			0.2		mV
Input referred noise			50		nV(rms)
Temperature drift	Input offset		± 5		nV/°C
	Gain		± 5		ppm/°C
Input common mode rejection			100		dB
Power supply rejection			100		dB
Power supply voltage	DVDD	2.6		5.5	V
	AVDD	2.6		5.5	V
Analog supply current	Normal		1100		μ A
	Power down		0.3		
Digital supply current	Normal		100		μ A
	Power down		0.2		

(1) Settling time refers to the time from power up, reset, input channel change and gain change to valid stable output data.

Table 2 Key Electrical Characteristics

Analog Input

The differential input is designed to interface directly with a bridge sensor's differential output. It has a fixed gain of 128. The large gains are needed to accommodate the small output signal from the sensor. When a 5V reference is used at the VREF pin, the full-scale differential input voltage range is $\pm 20\text{mV}$.

Power Supply Options

Digital power supply (DVDD) should be the same power supply as the MCU power supply.

Analog power supply (AVDD) should not be higher than the digital supply (DVDD).

A/D conversion reference voltage (VREF) should be connected to loadcell's supply voltage. It can be connected directly to AVDD or through a resistor to reduce the power consumption by the loadcell.

Clock Source, Output Data Rate and Format

HX710 uses the on-chip oscillator as clock source. The nominal output data rate is 10 or 80SPS.

The output 24 bits of data is in 2's complement format. When input differential signal goes out of the 24 bit range, the output data will be saturated at 800000h (MIN) or 7FFFFFFh (MAX), until the input signal comes back to the input range.

Serial Interface

Pin PD_SCK and DOUT are used for data retrieval, input selection, output data rate selection and power down controls.

When output data is not ready for retrieval, digital output pin DOUT is high. Serial data rate is 4.7(s)3.1i2tstu

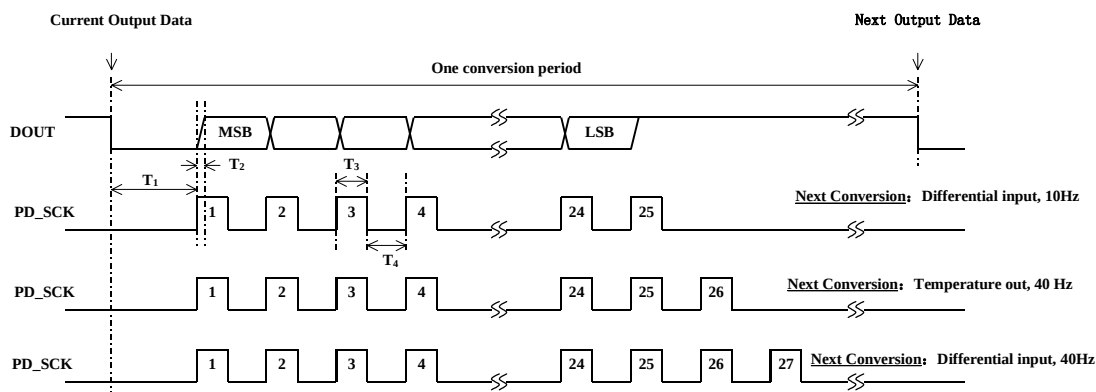


Fig.2 Data output, input and data rate selection timing and control

Symbol	Note	MIN	TYP	MAX	Unit
T ₁	DOUT falling edge to PD_SCK rising edge	0.1			μs
T ₂	PD_SCK rising edge to DOUT data ready			0.1	μs
T ₃	PD_SCK high time	0.2	1	50	μs
T ₄	PD_SCK low time	0.2	1		μs

Reset and Power-Down

When chip is powered up, on-chip power on rest circuitry will reset the chip.

Pin PD_SCK input is used to power down the HX710. When PD_SCK Input is low, chip is in normal working mode.

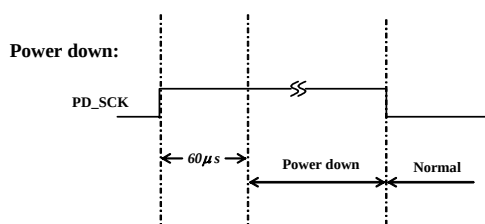


Fig.3 Power down control

When PD_SCK pin changes from low to high and stays at high for longer than 60 μs, HX710 enters power down mode (Fig.3). When PD_SCK returns to low, chip will reset and enter normal operation mode.

After a reset or power-down event, input selection is default to differential input with 10 Hz output data rate.

Application Example

Fig.1 is a typical weigh scale application using HX710.

Reference Driver (Assembly)

```

/*-----
Call from ASM:      LCALL   ReaAD
Call from C:      extern unsigned long ReadAD(void);
                    .
                    .
                    unsigned long data;
                    data=ReadAD();
                    .
                    .
-----*/

PUBLIC      ReadAD
HX710ROM    segment   code
rseg       HX710ROM

sbit       ADD0 = P1.5;
sbit       ADSK = P0.0;
/*-----
OUT:      R4, R5, R6, R7   R7=>LSB
-----*/

ReadAD:
    CLR     ADSK           //AD Enable (PD_SCK set low)
    SETB    ADD0           //Enable 51CPU I/O
    JB      ADD0,$         //AD conversion completed?
    MOV     R4,#24

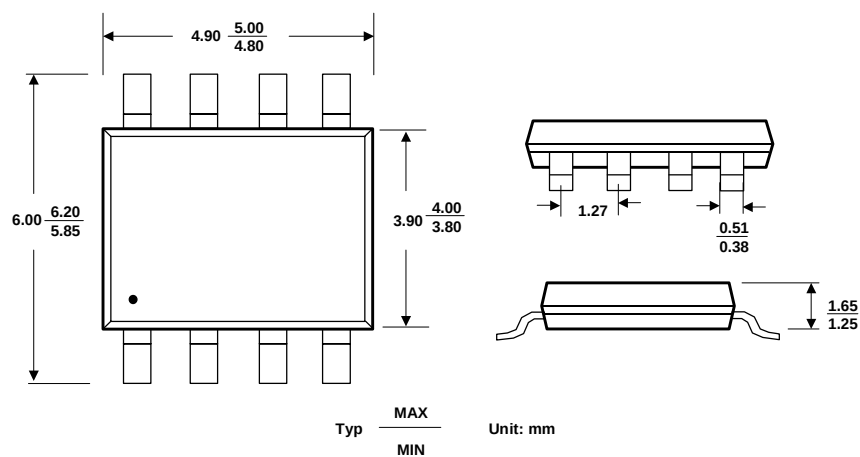
ShiftOut:
    SETB    ADSK           //PD_SCK set high (positive pulse)
    NOP
    CLR     ADSK           //PD_SCK set low
    MOV     C,ADD0         //read on bit
    XCH     A,R7           //move data
    RLC     A
    XCH     A,R7
    XCH     A,R6
    RLC     A
    XCH     A,R6
    XCH     A,R5
    RLC     A
    XCH     A,R5
    DJNZ    R4,ShiftOut    //moved 24BIT?
    SETB    ADSK
    NOP
    CLR     ADSK
    RET
END

```

Reference Driver (C)

```
//-----  
sbit  ADD0 = P1^5;  
sbit  ADSK = P0^0;  
unsigned long ReadCount(void) {  
    unsigned long Count;  
    unsigned char i;  
    ADD0=1;  
    ADSK=0;  
    Count=0;  
    while(ADD0);  
    for (i=0;i<24;i++) {  
        ADSK=1;  
        Count=Count<<1;  
        ADSK=0;  
        if(ADD0) Count++;  
    }  
    ADSK=1;  
    Count=Count^0x800000;  
    ADSK=0;  
    return(Count);  
}
```

Package Dimensions



SOP-8 Package